A Which lever? (Study 1)

Target selection
bring into view an object one cannot yet name

Context selection
fill the view with referents already ruled out

Mutual exclusivity
a learned word's referent is excluded for all others

Bottleneck: rare words are heard too seldom

Bottleneck: frequent competitors are almost always in view

Few referents in view ($C = 10$): target selection does most of the work (19×). Many ($C = 100$): learning never finishes unless the competitor bottleneck is removed; mutual exclusivity removes it but leaves target selection idle.

B How much, and over what? (Study 2)

Some of the time: 1 episode in 4 is chosen

captures much of the benefit; for the guess-test learner, partial control can beat full control

Among what is at hand: a choice set of k words

drawn by frequency: {ball, cup, dog, car, hat, . . . } common words

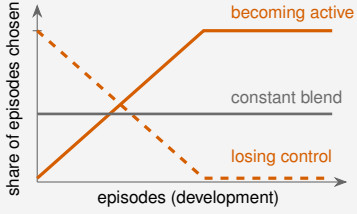
drawn uniformly: {kite, cup, owl, yak, dog, . . . } rare words too

Over-representing frequent words: up to 11× slower; uniform sets cost almost nothing. Goldilocks preference helps by spreading targets.

Objects or words: seek unnamed objects, or only words already heard

3.4× rather than 19× faster than passive ($C = 10$); without passive input, can fixate on one word.

C When? (Study 3)



Becoming active late is up to 15% faster than the same share of control spread evenly.

Losing control before the vocabulary is learned is up to 2.3× slower.

D At what cost? (Study 4)

photo
"a dog runs on grass"
"a brown dog plays"
3 more captions

move on
(+ search cost)

new photo
built around another word

stay: hear another caption

Moving on wins when seeking a new photo is cheap; staying wins once it costs about one or two extra episodes (up to 7% faster for the eliminative learner, 28% for the guess-test learner).